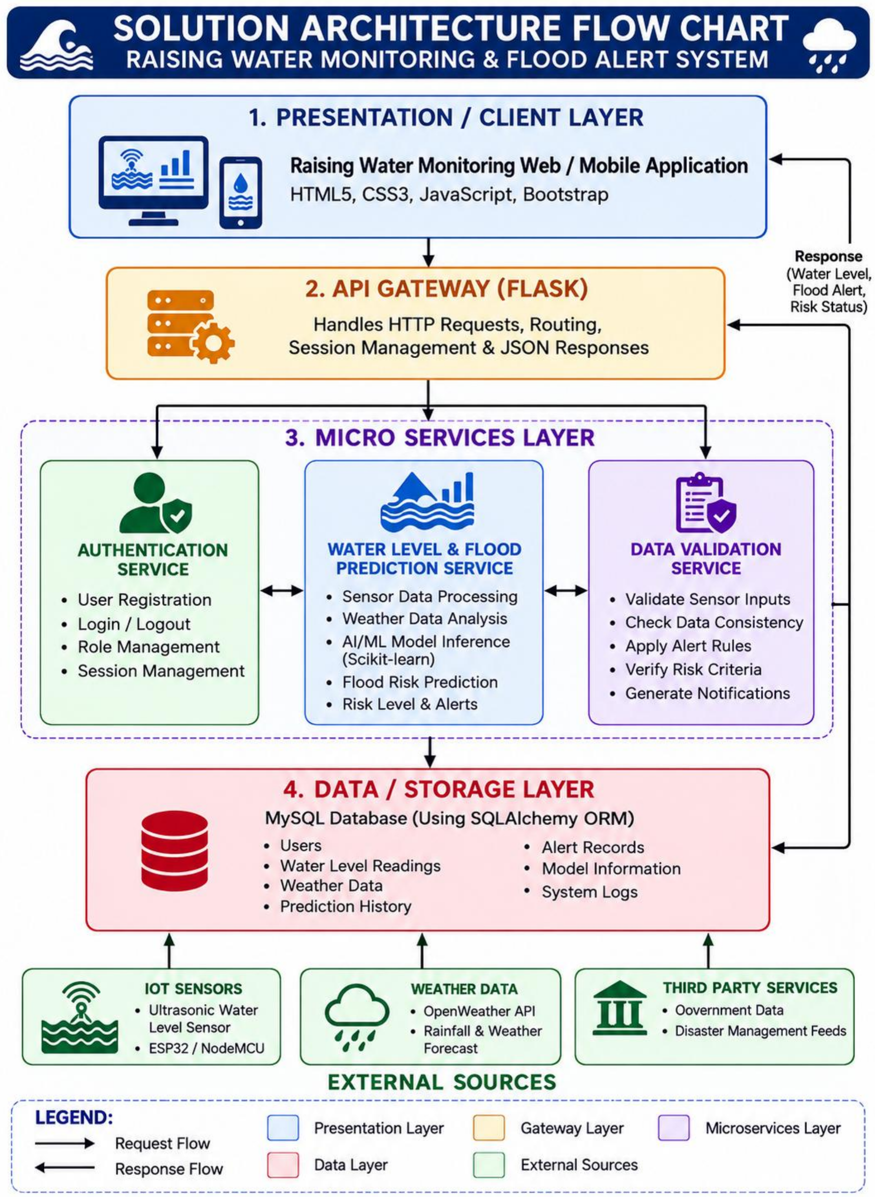
Solution Architecture

Date	30 June 2026
Team ID	
Project Name	Rising Waters
Maximum Marks	5 Marks

Solution Architecture Diagram Component Description Table



COMPONENT DESCRIPTION TABLE – RAISING WATER MONITORING & FLOOD ALERT SYSTEM

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides the user interface where users can view real-time water levels, flood alerts, risk status, maps and reports.	HTML5, CSS3, JavaScript, Bootstrap
API Gateway	Receives client requests from web/mobile application, processes REST APIs, routes requests to appropriate services and controls user sessions.	Flask
Authentication Service	Handles user registration, login, authentication, role-based access control and session management securely.	Python, Flask-Login
Water Level & Flood Prediction Service	Uses AI/ML models to analyze sensor and weather data, predict flood risk levels and generate alerts.	Python, Scikit-learn
Data Validation Service	Validates sensor data, weather data and user inputs (e.g., thresholds, location), checks data consistency and risk criteria before storing or generating alerts.	Python, Pandas
Database	Stores users, sensor data, weather data, alert records, risk levels, logs, and model-related information securely.	MySQL, SQLAlchemy ORM